

Modeling and Design for an Electrooptic Directional Coupler Modulator

Ahmed A. Abou El-fadl, Member, IEEE and Ahmed M. J. Dabroom
Jeddah College of Telecom. & Electronics, P.O.Box 2816, Jeddah 21461, Saudi Arabia
ahmedabouelfadl@hotmail.com, amdabroom@yahoo.com

Abstract — A numerical model is presented for an electrooptic directional coupler modulator. The model depends on the analysis of simple coplanar uniform electrodes. The analysis is based on the transverse resonance technique. An equivalent circuit is deduced to represent the structure of the modulator. General trial quantities are introduced as virtual adjustable sources in the equivalent circuit. It was observed that it is possible for such a device to be optimized by the variation of a number of fabrication parameters. The applied field and the change in optical waveguides optical field due to the antisymmetric refractive index change are calculated for the structure. Finally, the product of switching voltage and coupling length and optical loss due to the metal electrodes are also investigated

Index Terms — Coupling length, electrooptic directional coupler modulator, , optical loss, switching voltage, transverse resonance technique, trial quantities .

I. INTRODUCTION

The technology of choice for commercial 10-Gb/s and 40-Gb/s long-haul transmission systems are the LiNbO₃ - based electrooptic modulators [1]. LiNbO₃ modulators offer broad-band operation [2],[3], minimal and/or controllable chirp [3]–[5], long-term stability, [5] and have the added advantage that one design can be used for any one of a large number of transmission wavelengths. Directional couplers (DC's) are, in fact, the basis for several guided wave devices and are mainly used for optical switching networks. However, they can be used as both an intensity modulator or a switch. In the terminology of DC's there is a range of devices, such as the directional coupler with uniform electrodes [6], also called the switched DC [7], the reversed- DC switch [7], or that where the applied field produces changes in the coupling coefficient, [8], which in some cases has been called a DC modulator [9]–[10].

In this paper, a numerical model is presented for an electrooptic DC modulator with a simple uniform electrodes. This type of structure operates on the principle that the applied modulating field changes the refractive index in the two guides, such that the change is antisymmetric, which then affects the light wave propagation in these two guides, the coupling length and the phase matching between them, and finally, in turn, affects the power transfer efficiency.

The analysis is based on the transverse resonance technique. An equivalent circuit is deduced to represent the structure of the modulator. General trial quantities are introduced as virtual adjustable sources in the equivalent circuit [11]. It was observed that it is possible for such a device to be optimized by the variation of a number of fabrication parameters. The transverse operator method is used to find the optical and coupling characteristics of the DC without applied voltage [12]–[14]. The optical properties of the DC modulator are investigated through the linking between the full wave hybrid-mode analysis (transverse resonance technique) [11],[15] and the transverse operator method [12]–[14].

II. FORMULATION OF THE PROBLEM

A. Electrooptic directional coupler modulator structure

The schematic for electro-optic directional coupler modulator considered is shown in Fig.1

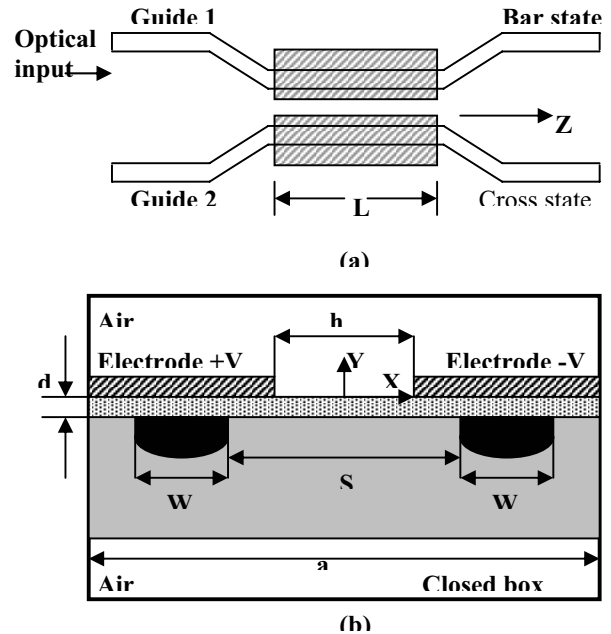


Fig.1 Schematic of DC modulator: (a) layout and (b) cross section.

B. The transverse operator method TOM

The propagation properties of the DC in Fig.1 without applied voltage can be characterized by the transverse electromagnetic components. Assuming that the propagation is in the Z-direction, the following relationship can be deduced from Maxwell's equation[12]-[14]

$$L' \cdot E_t(x, y) = K_z^2 E_t(x, y) \quad (1)$$

with

$$L' = (K_0^2 \epsilon_r + \nabla_t^2) \bar{I} + \left[\frac{\partial}{\partial x} \left[\frac{(\partial \epsilon_r / \partial x)}{\epsilon_r} \right] \quad \frac{(\partial \epsilon_r / \partial y)}{\epsilon_r} \right] \quad (2)$$

K_z is the propagation constant, K_0 is the free space propagation constant, $\epsilon_r(x, y)$ is the relative permittivity, and ∇_t^2 is the transverse Laplace operator.

The multi-dielectric structure of the DC is enclosed in a metallic box[12]-[14]. No loss of generality will be introduced, even if we treat an open structure, the surface wave nature of the propagating modes in the operating frequency ranges is considered. Furthermore, since the electric fields, as well as the magnetic field, decrease rapidly beyond the dielectric core, one can choose either electric or magnetic wall without affecting the numerical results. The relative permittivity can then be represented by a function of (x,y) co-ordinates, discontinued at the dielectric interfaces as[12]-[14]

$$\epsilon_r(x, y) = \sum_{i=1}^{n_y} \sum_{j=1}^{n_x} \epsilon_{ij} UX_{j,j-1} UY_{i,i-1} \quad (3)$$

with

$$UX_{j,j-1} = U(x - x_{j-1}) - U(x - x_j)$$

$$UY_{i,i-1} = U(y - y_{i-1}) - U(y - y_i)$$

$U(X)$ being the Heaviside function. The Delta-Dirac function, δ_a , will appear during the derivation of this function. The transverse operator L' given in (2) can be easily developed by replacing ϵ_r and its derivatives. Since (3) is a general form, L' has been obtained for the structure. Once the transverse L' is obtained, (1) can be solved by applying Galerkin's approach as:

- 1- the transverse electric field E_t is expanded in terms of chosen basis functions e_{tn} of amplitudes a_n as

$$E_t = \sum_n a_n e_{tn} \quad (4)$$

- 2- E_t in (1) is replaced by its expression in (4) and the inner products taken with each basis function e_{tn} .
- 3- the following matrix equation can be deduced as:

$$[L_{mn}][a_n] = K_z^2 [N_{mn}][a_n] \quad (5)$$

For a given Frequency, the solution of (5) produces both the even and odd propagation constants and the corresponding field components.

The choice of the basis functions may be arbitrary but the eigenmode of a corresponding waveguide is preferred which is of a similar nature to that the DC structure to be studied [12]-[14]. The dielectric stratified waveguide which is one of the corresponding DC structure provides a better convergence because the surface can only exist in the stratified structure. The TE-x and TM-x modes of this structure can be calculated by a transfer matrix formulation [16].

C. Optical properties of DC modulator

The DC modulator shown in Fig.1 modulator works on the principle that when no voltage is applied, the two guides are phase-matched and light input into guide 1 is nearly 100% coupled into guide 2 (cross-state); hence, the length over which the light couples L may be defined as

$$L = L_{co} = \frac{\pi}{\Delta\beta} \quad (6)$$

where $\Delta\beta = \beta_e - \beta_o$

L the interaction length of the device;

L_{co} the coupling length, when no modulating voltage is applied;

β_e the propagation constant for even mode;

β_o the propagation constant for even mode.

β_e and β_o can be calculated from the solution of (5) and also the optical fields.

D. Calculation of the applied electric field

When a voltage is applied as in Fig.1, there is a change in the refractive index due to the applied electric field which can be made unequal in the two guides by implementing an appropriate design. This leads to a mismatch in the phase velocities in both guides and also to

a reduction in L_c . By adjusting the device parameters most of the power can be transferred back to guide 1 (the bar-state where the coupling length, L_c , is given by $L_c = L_{co}/2$ at the end of the DC.

Although Δn_{ij} is defined as being equal to $r_{ij}E_j$ [5], it is well known that r_{33} the component is the largest and that the component of the modulating field E in the direction of the optic axis, E_y causes the largest change due to the linear electrooptic effect. Therefore, the expression for the change in refractive index with applied electric field is given by the linear electrooptic effect.

$$\Delta n(x, y, V) = -\frac{n^3}{2} r_{33} E_y(x, y) \quad (7)$$

The crystal cut is important in device fabrication. As mentioned above, the maximum electrooptic effect is only achieved when the electric field is parallel to the optical axis Z as in Fig. 1 and the polarization of the light is also in the same direction. However, for the DC modulator, the LiNbO₃ has to be Z-cut and hence, the optical field polarization is TM-x mode.

The applied electric field $E_y(x, y)$ can be calculated by using the transverse resonance technique and Maxwell's equations as in [17]. The equivalent circuit in Fig.2 is considered when the applied electric field has been calculated.

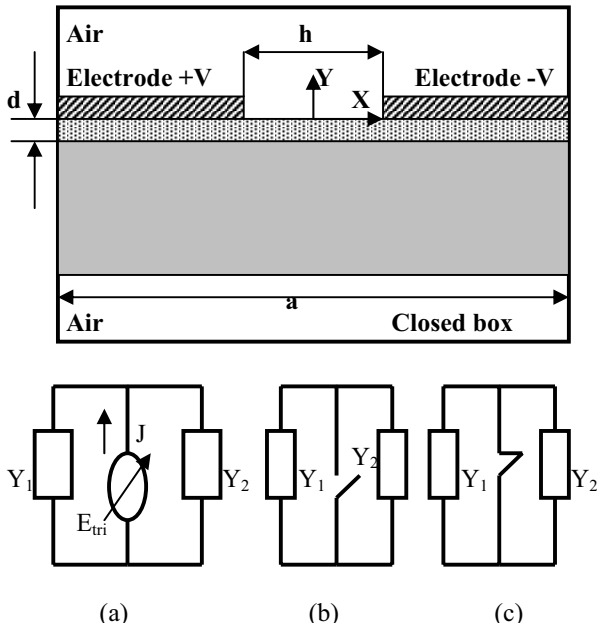


Fig.2 Cross section of the slot line with (a) its equivalent circuit in slot and under electrodes, (b) Equivalent circuit in slot and (c) equivalent circuit under the electrodes.

Finally, (5) is again solved for the two waveguides of the DC under an applied modulation voltage with the new refractive index profile

$$n_{\text{new}}(x, y, v) = n_{\text{old}}(x, y) + \Delta n(x, y, v) \quad (8)$$

in order to obtain the coupling length for a given modulating voltage. The voltage at which the coupling length is half that without the applied potential, due to an additional radian phase shift, is known as V_π or the switching voltage, and in this case the majority of the output power will be output from guide 1. The product of $L_{co} \cdot V_\pi$ has been estimated for various fabrication parameters

E. Optical loss and electrode design

Electrode design is an important issue in order to maximize the overlap between the optical and electric fields. The overlap can be optimized by varying a number of fabrication parameters, including buffer layer thickness. However, due to the decay of the electric field in the buffer and the Ti : LiNbO₃ material there is, in fact, only a small overlap between the optical and the electric field. Nevertheless, the buffer layer is needed in order to reduce the optical loss due to the lossy metal electrodes. The perturbation method is used in conjunction with the transverse resonance technique to calculate the optical loss due to the metal electrodes.

III. NUMERICAL RESULTS

A Ti-indiffused Z-cut Y-propagating weakly coupled DC modulator with guide width, W , of $9\mu\text{m}$, separation between the guides, S , of $7.5\mu\text{m}$, and electrode separation, h , of $5\mu\text{m}$, is first considered. The SiO₂ buffer layer has a thickness, d , which is $0.2\mu\text{m}$. At the operating wavelength of $1.56\mu\text{m}$, the extraordinary and ordinary refractive indices for LiNbO₃, are taken to be $n_e = 2.14$ and $n_o = 2.22$, the change in the refractive index due to the Ti indiffusion is 0.01 for the extraordinary index, and 0.005 for the ordinary index.

Fig.3 illustrates the relationship between the coupling length L_c and the applied voltage with the separation between guides S as a varying parameter. For each separation the coupling length is decreased with increasing the applied voltage.

From Fig.3 the switching voltage V_π is 25.35 volt for the smaller separation S of $5\mu\text{m}$ and is 7.68 volt for larger separation S of $7.5\mu\text{m}$.

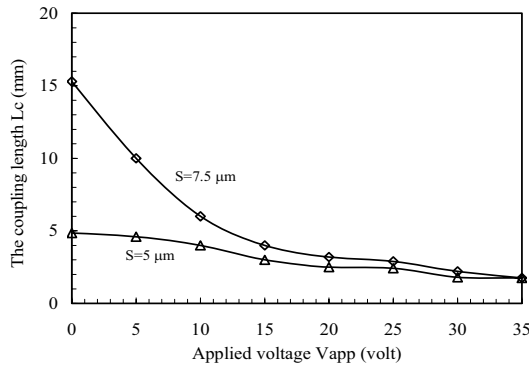


Fig.3 Dependence of the coupling length L_c on the applied voltage for the directional coupler.

Dependence of coupling length L_{co} , switching voltage V_π and the product of coupling length and switching voltage $L_{co}.V_\pi$ on waveguides separation S is shown in Fig.4. However, as the switching voltage needs to be at a more practical value (i.e., ≤ 10 V), a higher separation is desired. From Fig.4 a separation of $7.5\mu m$ is an appropriate value.

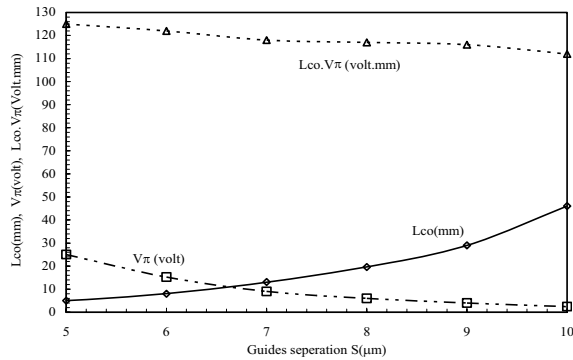


Fig.4 Coupling length L_{co} , switching voltage V_π and the product of coupling length and switching voltage $L_{co}.V_\pi$ versus waveguide separation S .

The buffer layer thickness, d , influences the overlap between the modulating and optical fields. Dependence of coupling length L_{co} , switching voltage V_π , and the product of coupling length and switching voltage on the buffer layer thickness d is shown in Fig.5

The variation of L_{co} is very small with the buffer thickness, d , but the switching voltage V_π shows a sizable increase, which is understandable because as the buffer layer thickness increases, the optical mode size increases as observed in Fig.5, hence V_π has to increase since the switching voltage is minimized for small optical modes.

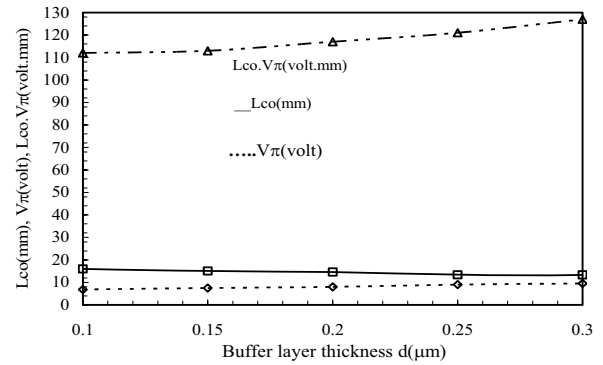


Fig. 5 Coupling length L_{co} , switching voltage V_π and the product of coupling length and switching voltage $L_{co}.V_\pi$ versus buffer layer thickness d .

The device performance for different values of h has been calculated in terms of V_π is shown in Fig.6. This figure illustrates that V_π increases with increasing h .

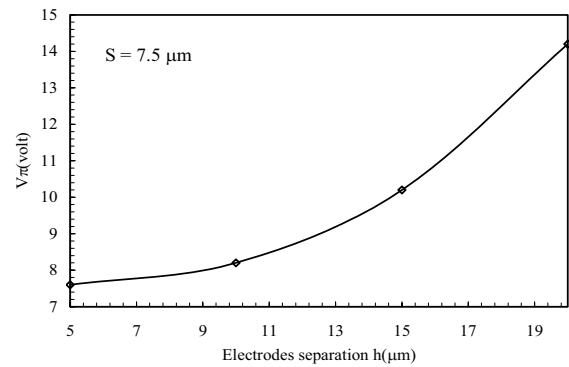


Fig.6 Switching voltage V_π versus of electrodes separation h .

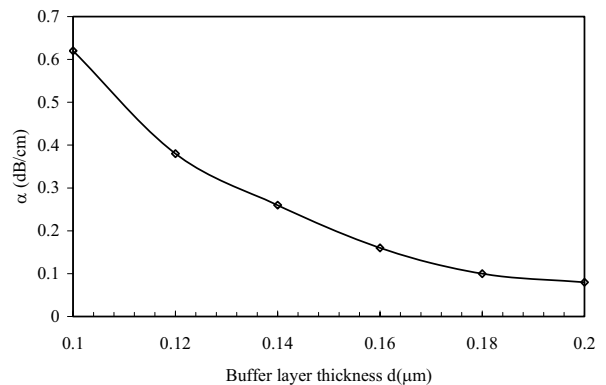


Fig.7 The optical loss due to the electrode α versus the buffer layer thickness d for the gold electrodes with thickness= $0.15\mu m$.

The model loss $\propto$ for TM-x polarized mode is shown as in Fig.7 for gold electrodes for an electrode thickness of 0.15 μm .

From Fig.7, one can be noticed that when the buffer layer is increased the loss is decreased. So a thicker buffer layer is needed in order to reduce the absorption loss by shifting the modal field away from the metal electrodes], but, as shown earlier in Fig. 5, an increased buffer layer thickness also means a higher product $L_{co}.V\pi$.

V. CONCLUSION

A numerical model is presented for an electrooptic directional coupler modulator. The model depends on the analysis of simple coplanar uniform electrodes. The analysis is based on the transverse resonance technique. An equivalent circuit is deduced to represent the structure of the modulator. General trial quantities are introduced as virtual adjustable sources in the equivalent circuit. It was observed that it is possible for such a device to be optimized by the variation of a number of fabrication parameters. The applied field and the change in optical waveguides optical field due to the antisymmetric refractive index change are calculated for the structure. Finally, the product of switching voltage and coupling length and optical loss due to the metal electrodes are also investigated.

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